

Response to the Independent R45 Referee Report: R47

Anonymous manuscript for Operations Research

September 25, 2026

Version and substantive revision

This response addresses the independent R45 report dated September 25, 2026, at review commit 260a5224c55e0326d9da7f0c813b4a3fcab7671f. The report reviewed manuscript 9c52e2d58d69a8d653c3b50e2c860b35db04dc7f. We have retained the subsequent R46 development at 9f260e0d6c4e06cf4227f96865a88f4dab67e58e and revised it on the separate R47 branch. We distinguish new R47 claims from repairs already made in R46. The new contribution is certified response coarsening for distinct histories, its original-contract lift, an additive scheme with variable history count, and monotone refinement. No result below changes the original command budget, charges, exact promise, or individual realization constraints. All historical sources remain intact, and the predecessor readers are preserved byte for byte.

12.1. Complexity frontier: a positive variable-history result

The report correctly identifies that the old target net does not explain the role of history count. New Theorem 7.3 supplies an alternative to a hardness claim. For rational quadratic primitives, bounded reward and shortfall-curvature scales, a fixed number of eligible catalog prefixes, and any fixed positive additive accuracy, joint finite-catalog design is polynomial in the number of histories, catalog size, command budget, and rational input length. The construction bins caps and curvatures, keeps eligibility exact, and isolates a minimum-cap anchor guard. Lemma 7.2 bounds the reduction error and the number of reduced groups independently of raw history count. The proof gives an explicit conservative dyadic node bound. This is not a fully polynomial approximation scheme: dependence on accuracy and eligibility complexity can be very large. No NP-hardness classification is asserted. The inherited fixed-budget enumeration, saturation, exact-type, and continuous-design regimes remain in the manuscript; the provenance table separates these from the four R47 statements.

12.2. Structural global method and feasible disaggregation

R46 supplied the two-sided price-path decomposition and adaptive target-box cover, replacing the full target grid as the main joint algorithm. R47 extends this mechanism beyond exactly repeated histories. Theorem 7.1 uses weighted mean caps and the minimum cost curvature within a common eligibility group to obtain an

optimistic joint problem. The mean-cap construction preserves total capacity, including saturated promises. It is not used as a substitute for original individual caps. A clip-and-fill lift preserves each group mean, restores each individual cap, and executes the same selected commands at the same opening charge. Its exact weighted transfer identity yields both a posterior error bound and a uniform bound. The theorem sandwiches the original optimum between a feasible original policy and the reduced upper bound. Proposition 7.4 proves monotonicity of the optimistic optima under refinement; extrema of all valid bounds give a nested interval even when individual solves stop early. A minimum-cap counterexample demonstrates why unrestricted averaging is invalid.

12.3. Nontrivial accuracy and documented limits

The new study executes 24 specified joint-design instances at absolute tolerance 0.001, with 4 through 256 distinct histories, 9 or 17 catalog levels, budgets two or three, one or four eligible prefixes, and charged as well as uncharged instances. All 24 coarsened policies obtain a checked original-space interval of width at most 0.001; the largest width is approximately 0.000783098. Each 256-history input has 256 distinct exact response types, not replicated branches. These are clustered synthetic perturbations and are not evidence that all populations admit five useful groups. The conservative accuracy bins in Theorem 7.3 can be much finer than the successful practical bins; practical completion is accepted only from the checked original-space interval. The inherited R46 grid study, including incomplete nets and their withheld accuracy guarantees, remains intact in the companion. The new claim is not that exhaustive nets became practical.

12.4. Failures, heuristic distinctions, and refinement

The R47 headline concerns joint upper and lower certificates, not target-portfolio exact-hit counts. A separate, disclosed supplementary protocol contains four deliberately dispersed three-history examples. In all four, the reduced problem is solved but the original requested tolerance is not met: initial gaps are 0.0421875, 0.044375, 0.11, and 0.67. Splitting the remaining mixed group produces exact original solutions in these four examples. The manuscript retains both stages, rather than discarding the unresolved coarse relaxations. These examples establish neither a universal one-split bound nor general heuristic exactness. The older portfolio results and the charged failure remain historical evidence, with their corrected sample-only labels. All 24 primary/scaling outcomes, all eight stress stages, and the corresponding exact certificates are retained.

12.5. Independent comparison and reproducibility

The 24 new cases compare coarsening, the uncoarsened R46 box method, and the inherited priced-prefix method under the same original objective and requested tolerance, with common nominal node and soft time limits. They complete 24, 19, and 21 cases respectively. Median measured local times are approximately 0.0375, 0.3183, and 0.1691 seconds. Soft time checks occur between oracle blocks and are documented rather than presented as hard deadlines. Every primary case also has an all-book joint reference; this reference

shares the exact allocator and is labeled accordingly. A separate uneliminated mixed-integer formulation is run with both quadratic envelopes on all 12 primary cases. All 24 solves return optimal numerical status; per-case time, nodes, numerical primal-dual gap, envelope error, bracket width, and residual information are saved. Floating-point MIP bounds are not rational certificates. Original local timings remain separate from subsequent publication replay timings. R46’s stronger priced-prefix outcomes and its four interrupted box cases remain in the companion.

12.6. Stronger certificates in the original history space

The reduced upper bound optimizes the shared command book and endogenous targets; it is not a Jensen bound at a prespecified target vector. The new posterior loss separates cap-restoration movement from cost dispersion, and charges cancel because the lift preserves the book. An independently implemented checker recomputes the representative data, guard, eligibility, error bounds, exact group masses, lower value, all original lottery constraints, and the inner Bellman/coverage certificate. It neither imports the coarsening optimizer nor trusts its completion flag. The test suite compares 72 exact original/reduced joint optima, 602 book-specific lifts, 441 cap-dispersion inequalities, 216 accuracy partitions, and 56 refinement comparisons. Eighteen semantic certificate corruptions are rejected. The older inexact-support theorem retains its explicit strong-oracle assumptions; it is not relabeled as an unconditional heuristic guarantee.

12.7. Operational interpretation and presentation

We answer the report through a stronger optimization theory rather than inventing field calibration. The introduction is rewritten around joint design, response coarsening, feasible recovery, and decision certificates. The original institutional and continuous-design contributions are retained; the previous R46 computational discussion moves intact to the companion. The current main reader has 39 nonreference pages and the companion has 39 pages. This meets the checked format bounds but is not a claim that the package is short or that the editorial length concern disappears. We preserve 45 antecedent mathematical statement labels and add four proved statements. The manuscript distinguishes writable command bits from the history encoder, individual cap table, probabilities, read-only constants, and certification metadata. The new Li–Bertsekas aggregation comparison and Fattahi managed-charging comparison identify adjacent literature, not application data for this paper. All study inputs are synthetic, all opening charges are supplied, and no deployment or empirical calibration is claimed.

Minor comments 1–4: implementation and coverage

1. The inherited single-history path uses one fixed-target optimization; the regression remains, and the new completed-scheme implementation also tests its single-group boundary. 2. Genuine clipping is now tested on unequal caps and weights at a saturated promise. The exact weighted lift distance in the recorded example is 3/20; individual caps, not an averaged replacement, are restored. 3. The independent full-cover checker

and the earlier lattice coverage audit are retained. Every new inner cover is checked separately, and mutation tests remove a leaf or corrupt its bound. 4. Runtime, catalog size, command budget, objective scaling, additive gap and scaled gap are saved for every new case. The old net timing and profile-growth comparison remains in the companion with incompleteness visible.

Minor comments 5–9: provenance and empirical wording

5. The charged saturated recurrence is identified as inherited from R42, rather than newly claimed in R47. 6. Certified joint design now means an upper bound on the joint original problem paired with a fully feasible original policy, including when coarsening is used. 7. The abstract explicitly limits the new variable-history result to bounded eligibility and primitive scales, and states that accuracy dependence can be large. The text retains the exponential type-dimension bound and gives the full new accuracy dependence. 8. The old safeguarded exact-hit column remains a sample observation, not a safe algorithmic guarantee. 9. The historical charged portfolio failure has exact value 1373/3600, core value 53/150, and loss 101/3600, about 7.36 percent of the exact value. This scale and the original counterexample are retained; R47 does not reinterpret that failure as success.

Minor comments 10–14: comparisons and theorem scope

10. Every direct MIP case includes both envelopes and the requested numerical solver and bracket fields, including the status field that would expose interruption; no case is omitted. 11. The old ideal-target inserted candidate pool remains an oracle sanity check, not a deployable global algorithm. 12. Conditional timing remains marked as targets fixed and is not used to support the R47 joint-completion claim. 13. The new abstract and theorem explicitly concern a finite catalog; they do not silently extend the coarsening scheme to unrestricted continuous commands. 14. The theorem provenance table distinguishes R42–R44 inherited results, R45 compression/grid results, R46 two-sided box results, and the four R47 coarsening/dispersion/scheme/refinement statements.

Minor comments 15–18: cardinality, scale, types, and numerical evidence

15. The budget counts selected commands. An exact-cardinality frontier may include a selected zero-charge level with no positive branch probability; this is not physical positive-use cardinality. 16. Additive tolerance is tied to the stated reward/cost normalization, with a reported objective scale; multiplying reward, costs and charges requires multiplying the tolerance as well. No multiplicative guarantee for potentially negative net values is asserted. 17. Exact response-type aggregation remains available, and R47 adds controlled coarsening of different caps and curvatures with exact eligibility. The new large-history examples expressly have no duplicate exact types. 18. Rational policy/upper-bound certificates and numerical MIP envelopes retain distinct labels. Neither floating-point solver status nor sample agreement is promoted into a proof.

Reader and evidence guide

Read `main.pdf`, `electronic_companion.pdf`, and this response. The R47 directory contains the four new statements and full proofs, fixed protocols, reproducible code, all case-level numerical records, compressed exact certificates, the independent checker, build validation, content map, and an inherited-tree preservation audit. The R46 predecessor readers remain byte-identical in predecessor/. Canonical experimental timings describe the original local runs. Publication reconstructs semantic certificates at their recorded node counts, checks their exact hashes, reruns tests separately, and compiles the readers on the isolated R47 branch. Main, review, R46 and other revision branches are not publication targets.